Name
Date
Course

Test IV

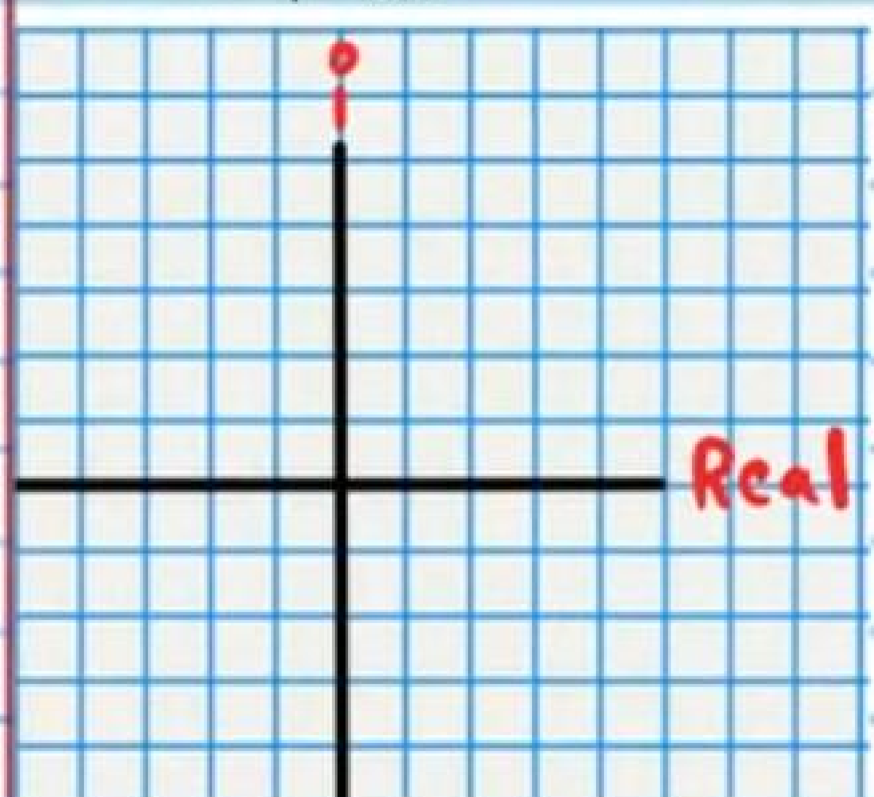
I) Write your name, the date, and the course title (i.e. Precalculus).
II) Show all work.
III) Draw a rectangle around your final answer for each problem.
IV) A graphing calculator is required.
V) Each problem is worth four points for a total of 100 points.

① Find the geometric mean of the data set below. Round to the nearest hundredth.

0.3, 1.1, 0.9, 0.77, 0.99, 0.5, 0.39

② i) Graph the following complex number.

$-1-2i$



ii) If "f" and "f^{-1}" are inverse functions, find $f\left(\frac{7}{2}\right)$ if $f^{-1}(20)=\frac{7}{2}$.

③ i) Use your calculator's base "10" logarithm button (LOG) and Logarithm Property IX to find the value of the expression below. Round to the nearest ten thousandth.

$\log_8 5$

ii) Given the following intensity of sound, find the loudness perceived by the human ear.

$10^{-5}\ W/m^2$

④ The population of a suspension of yeast after "t" hours can be predicted with the equation below.

$$P(t)=\frac{1{,}000}{1+21e^{-0.3t}}$$

i) What is the carrying capacity of the population?

ii) What is the initial population? Round to the nearest organism.

iii) What will the population be after 10 hours? Round to the nearest organism.

iv) If "f" and "g" are inverse functions, find $f(0)$ if $g(10)=0$.

⑤ Each table below represents a function "f." Write the values of the inverse function "f^{-1}." If "f" is not a one-to-one function, then you must omit values of "f^{-1}" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight or circle the omitted values.

i)

X	0	1	-9	81	80
f(X)	5	2	13	20	30

X					
$f^{-1}(x)$					

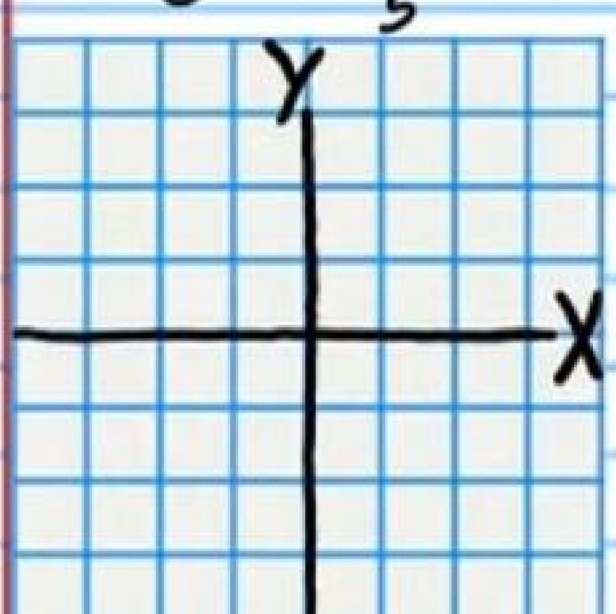
ii)

X	-8	3	2	15	0
f(X)	16	4	4	16	3

X					
$f^{-1}(x)$					

⑥ i) Graph the following polar equation.

$\theta = \frac{4\pi}{5}$



ii) Solve. Round to the nearest thousandth.

$8 = \frac{e^{3x}}{e^{7x}}$

⑦ For the function below, find an algebraic expression representing the difference quotient. Use "h" for "change in X." It is not necessary to simplify the expression.

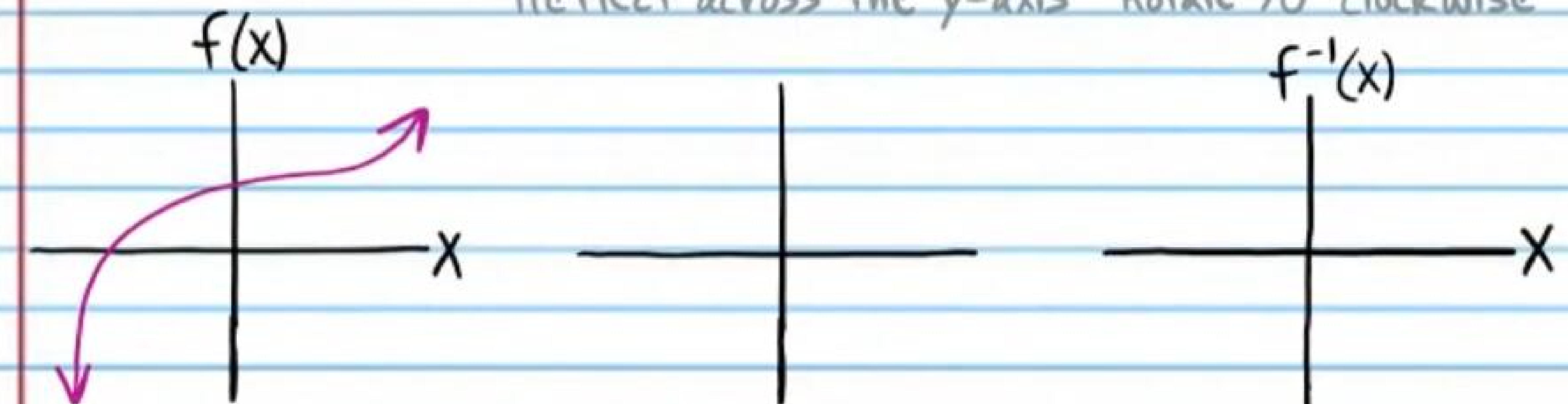
$f(x) = \ln(x+2)$

⑧ Given the graph of "f" below, draw the graph of the inverse function "f^{-1}."

Write the inverse function
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise



⑨ In 4.7 years, a radioactive substance decreases from 19.2 grams to 15.1 grams. How much of this same radioactive material would be left if we started with a 20 gram sample and monitored it for a year? Round to the nearest tenth.

⑩ The half life of Zinc-72 is 46.5 hours. If a docter has 10 grams of this substance, how much will remain after 30 hours? Round to the nearest hundredth.

(11) i) Abbey ran 43 miles during a three-day marathon. What was her average speed during the whole three-day marathon?

ii) During the marathon, Abbey stopped to sleep and eat. If she ran for a total of 1 day during the marathon, what was her average speed during the time that she was running?

(12) If a 2% interest rate is compounded continuously for 5 years, and the initial investment is \$2,635, how much would the investment be worth?

(13) Are the following functions inverses? Demonstrate algebraically.

$f(x) = \frac{4}{1-2x}$ & $g(x) = \frac{x-4}{2x}$

14) If $f(x) = 6x^2 + 13x - 1$ and $f(m) = 4$, find m.

15) Solve. Round to the nearest ten thousandth.

$$17 = \frac{10^{5x}}{10^{2x}}$$

Find an equation of the inverse of each function below.

16) $f(x) = 2x - 1$

17) $f(x) = 3\log_5(x-1) + 2$

(18) If $g(x)=5x^2+1$ and $f(x)=4x^2-2x$, find the coordinates of the point(s) where the graphs of these functions intersect.

(19) Solve.

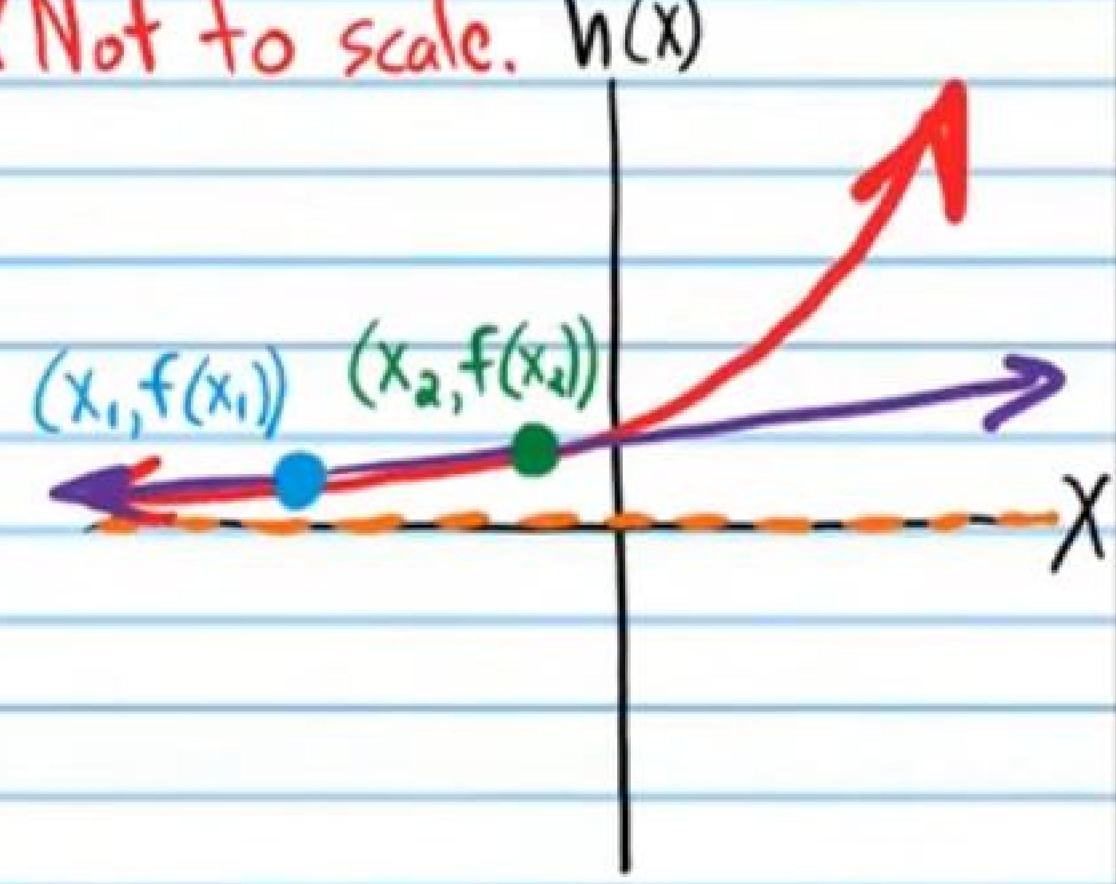
$$\frac{\log x}{2}+3=4$$

(20) Solve. Round to the nearest thousandth.

$$-4\ln x+10=9$$

(21) If $h(x)=3^x$, what is the average rate of change of the function per unit x from $x=-3$ to $x=-1$?

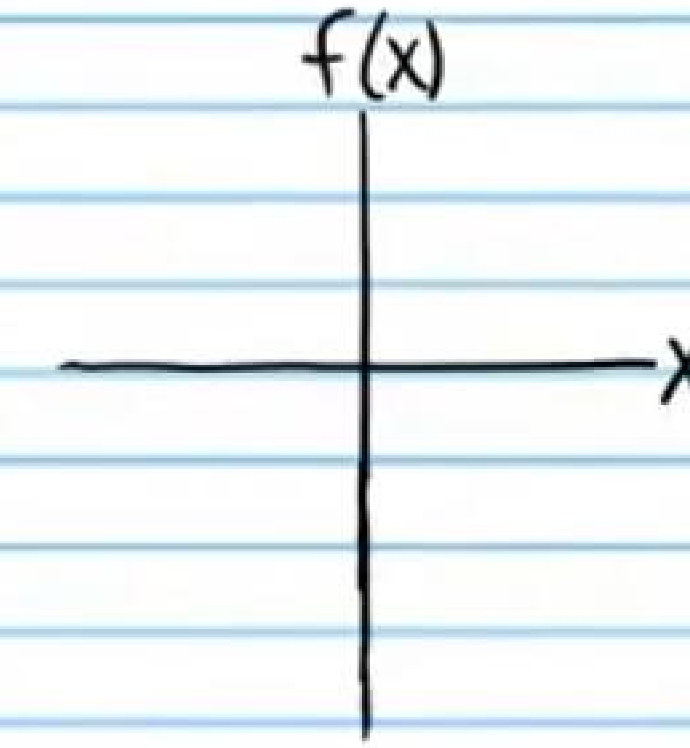
*Not to scale.

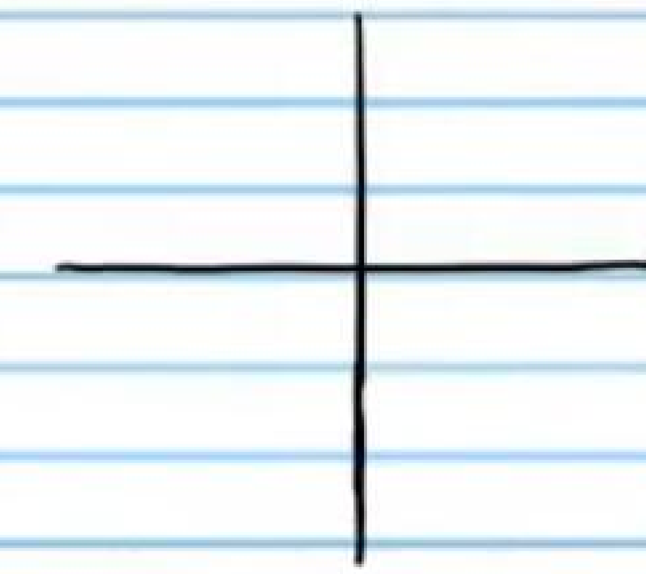


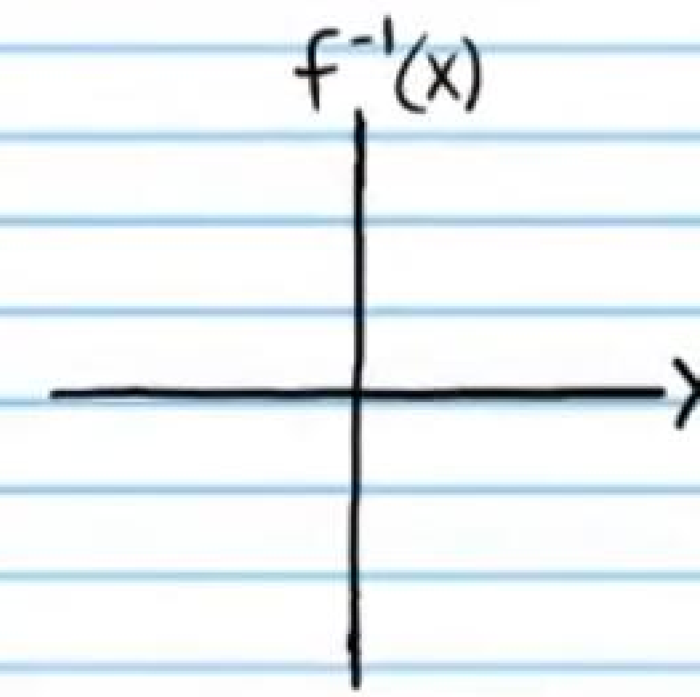
(22) If $f(x) = 2\sqrt{\frac{x}{3} - 5} + 6$ & $g(x) = 10$, for what value(s) of x does $f(x) \overset{?}{=} g(x)$

For problem #'s 23-24, find the equation of the inverse function. Restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

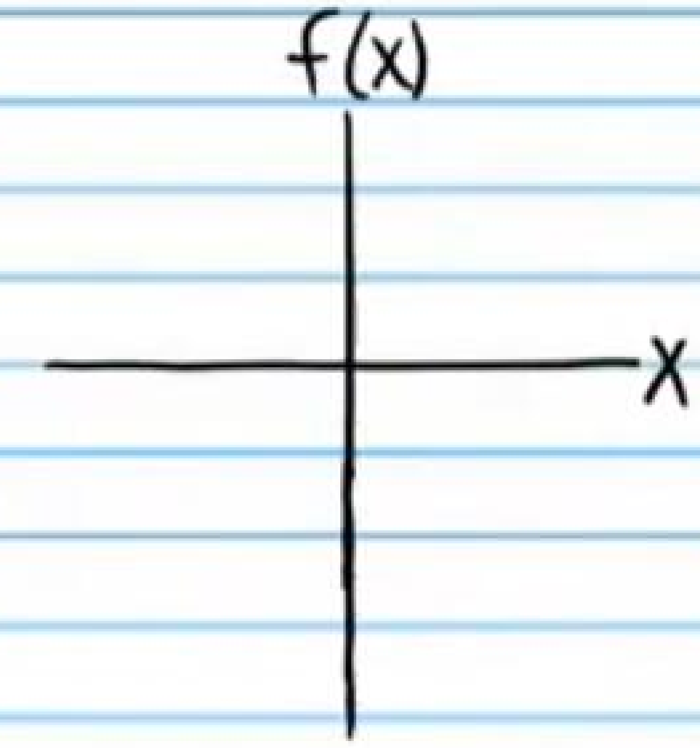
(23) $f(x) = (x+4)^2$

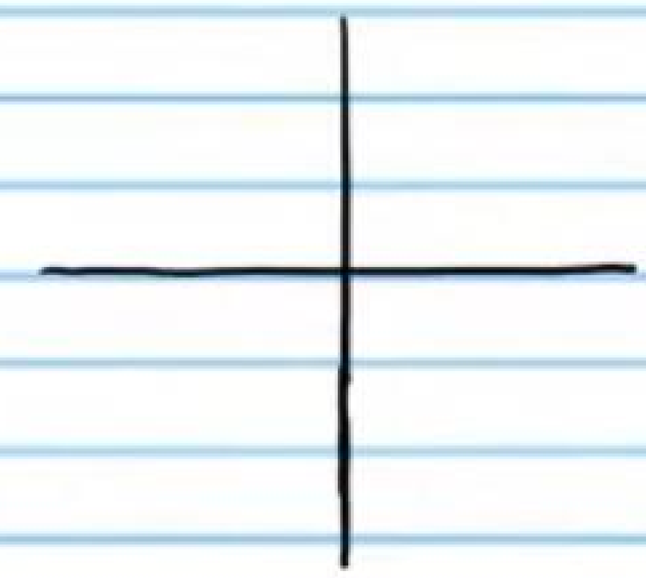


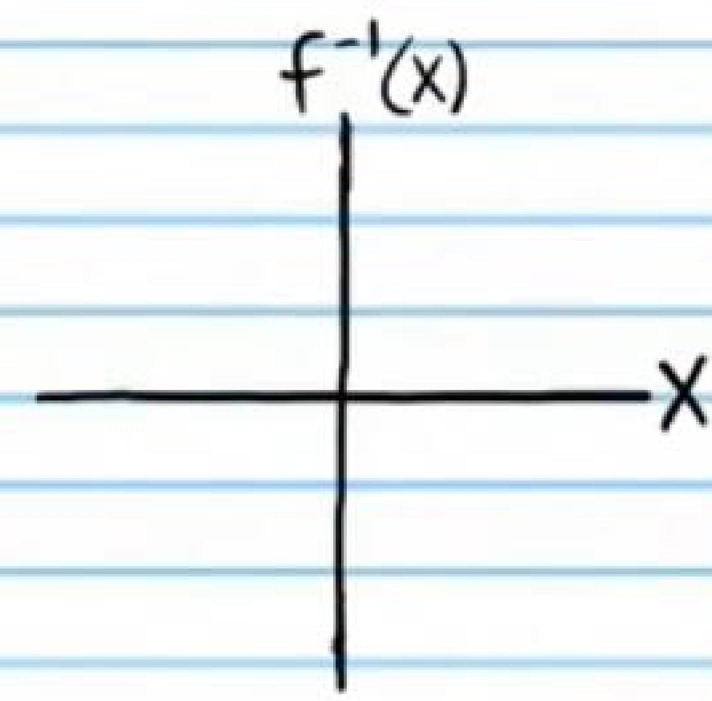




(24) $f(x) = \cos x$





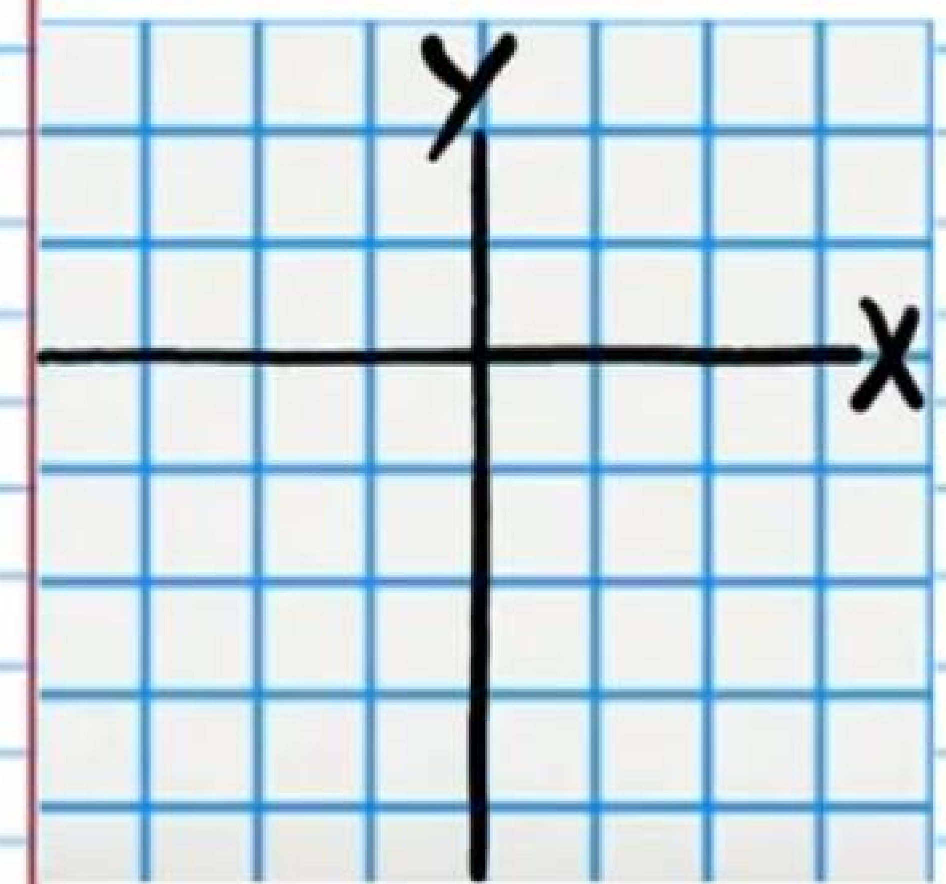


(25) Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$r = 1 - 3\sin\theta$

1st Point

2nd Point



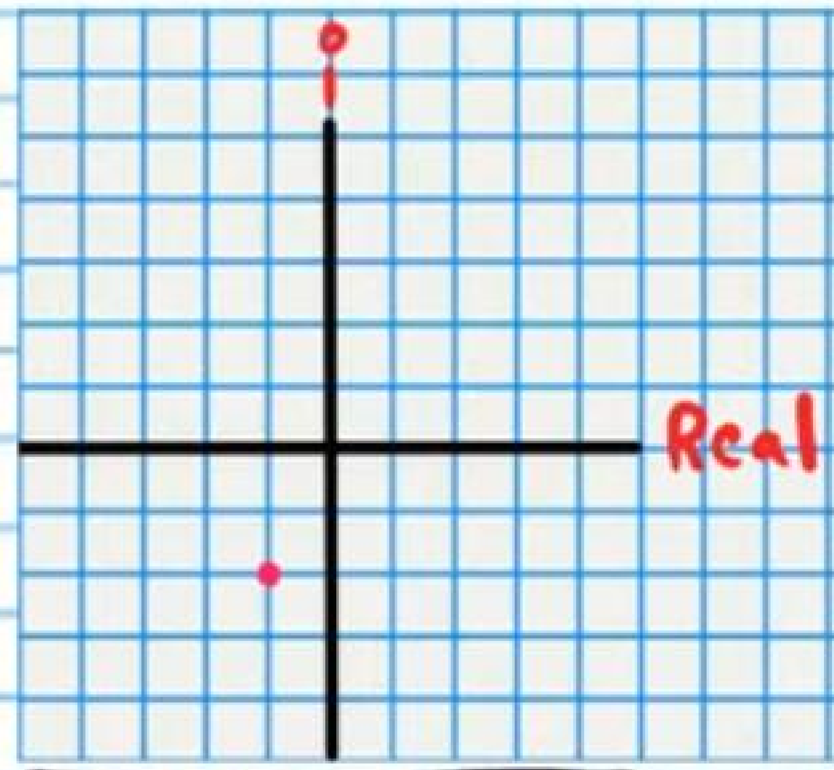
3rd Point

4th Point

Answers

① $\boxed{0.64}$

② i)



ii) $\boxed{f\left(\frac{7}{2}\right) = 20}$

③ i) $\boxed{\frac{\log 5}{\log 8} \approx 0.774}$ ii) $\boxed{70 \text{ dB}}$

④ i) $\boxed{1{,}000 \text{ organisms}}$ ii) $\boxed{45 \text{ organisms}}$ iii) $\boxed{489 \text{ organisms}}$

iv) $\boxed{f(0) = 10}$

⑤ i)

X	0	1	-9	81	80
f(X)	5	2	13	20	30

X	5	2	13	20	30
$f^{-1}(X)$	0	1	-9	81	80

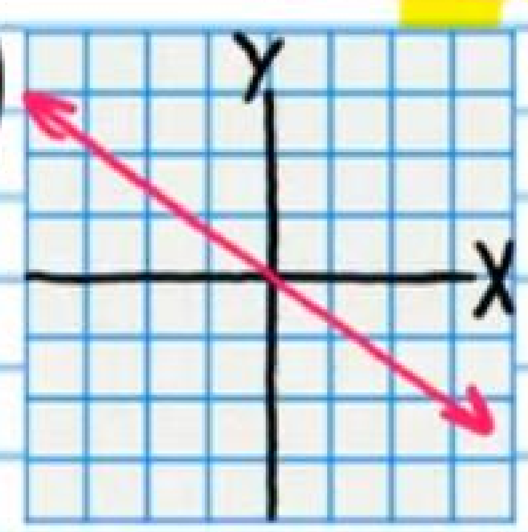
ii)

X	-8	3	2	15	0
f(X)	16	4	4	16	3

X	16	4	4	16	3
$f^{-1}(X)$	-8	3	2	15	0

Omit two of these four points.

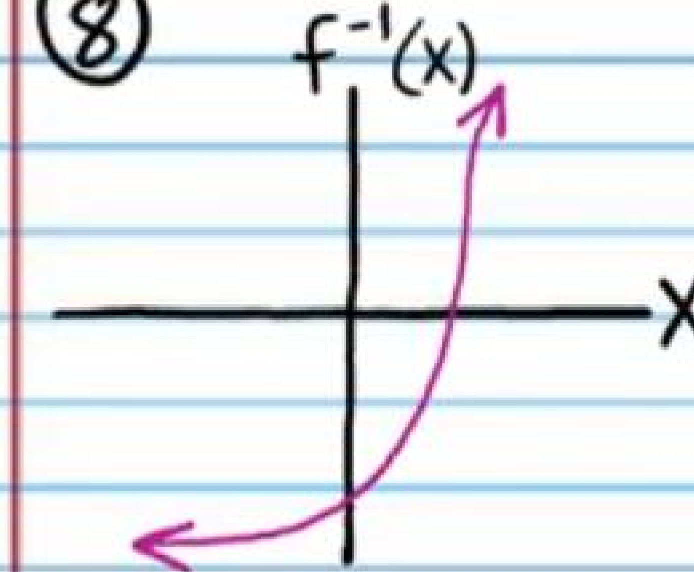
⑥ i)



ii) $\boxed{X \approx -0.52}$

⑦ $\boxed{DQ = \frac{\ln(x+h+2) - \ln(x+2)}{h}}$

⑧



⑨ $\boxed{19 \text{ grams}}$

⑩ $\boxed{6.39 \text{ grams zinc-72}}$

⑪ i) $\boxed{14\frac{1}{3} \text{ mi/day}}$ ii) $\boxed{43 \text{ mi/day}}$

⑫ $\boxed{\$\,2{,}912.13}$

⑬ $$f(g(x)) = \frac{4}{1-2\left(\frac{x-4}{2x}\right)} = \frac{4}{1-\frac{x-4}{x}} = \frac{4}{\frac{x}{x}-\frac{x-4}{x}} = \frac{4}{\frac{4}{x}}$$

$$= x \checkmark$$

$$g(f(x)) = \frac{\frac{4}{1-2x}-4}{2\left(\frac{4}{1-2x}\right)} = \frac{\frac{4}{1-2x}-4\frac{(1-2x)}{(1-2x)}}{2\left(\frac{4}{1-2x}\right)}$$

$$= \frac{\frac{4}{1-2x}-\frac{4-8x}{1-2x}}{2\left(\frac{4}{1-2x}\right)}$$

$$= \frac{\frac{8x}{1-2x}}{\frac{8}{1-2x}} = \frac{8x}{1-2x} \div \frac{8}{1-2x} = \frac{8x}{1-2x} \cdot \frac{1-2x}{8}$$

$$= x \checkmark$$

$\boxed{\text{Yes}}$

(14) $\boxed{M = \frac{1}{3} \text{ or } M = -\frac{5}{2}}$

(15) $\boxed{x \approx 0.4101}$

(16) $\boxed{f^{-1}(x) = \frac{x+1}{2}}$ or $\boxed{f^{-1}(x) = \frac{x}{2} + \frac{1}{2}}$

(17) $\boxed{f^{-1}(x) = 5^{\frac{x-2}{3}} + 1}$

(18) $\boxed{(-1, 6)}$

(19) $\boxed{x = 100}$

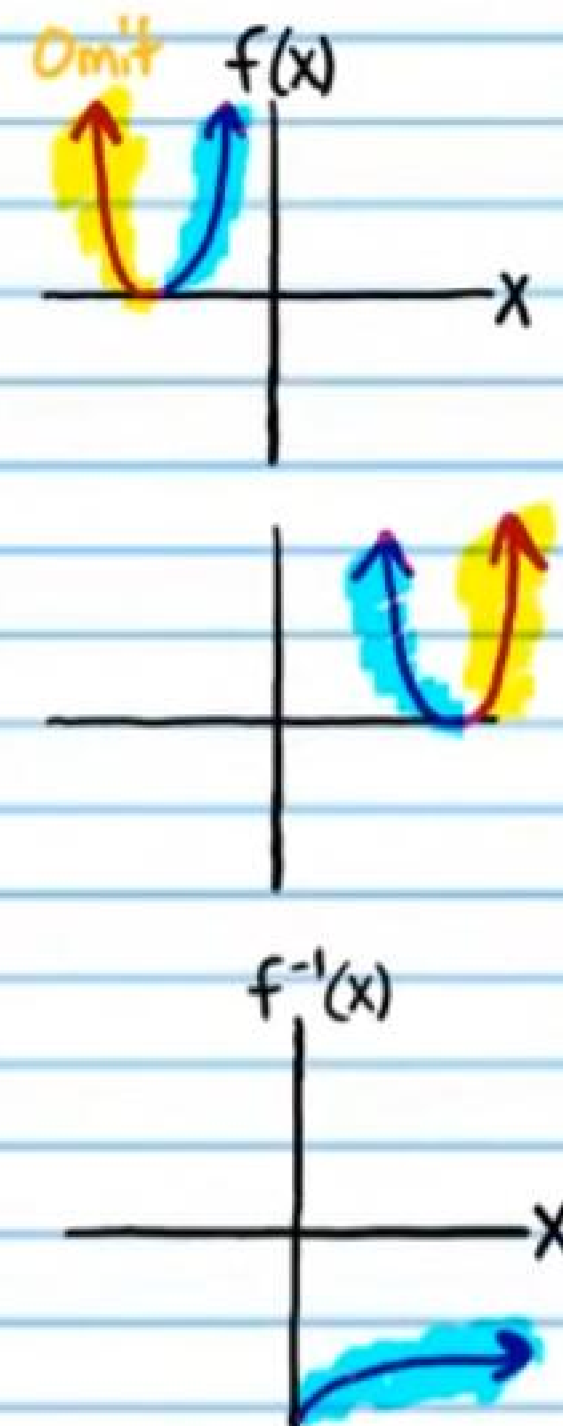
(20) $\boxed{x \approx 1.284}$

(21) $\boxed{\frac{4}{27}}$

(22) $\boxed{x = 27}$

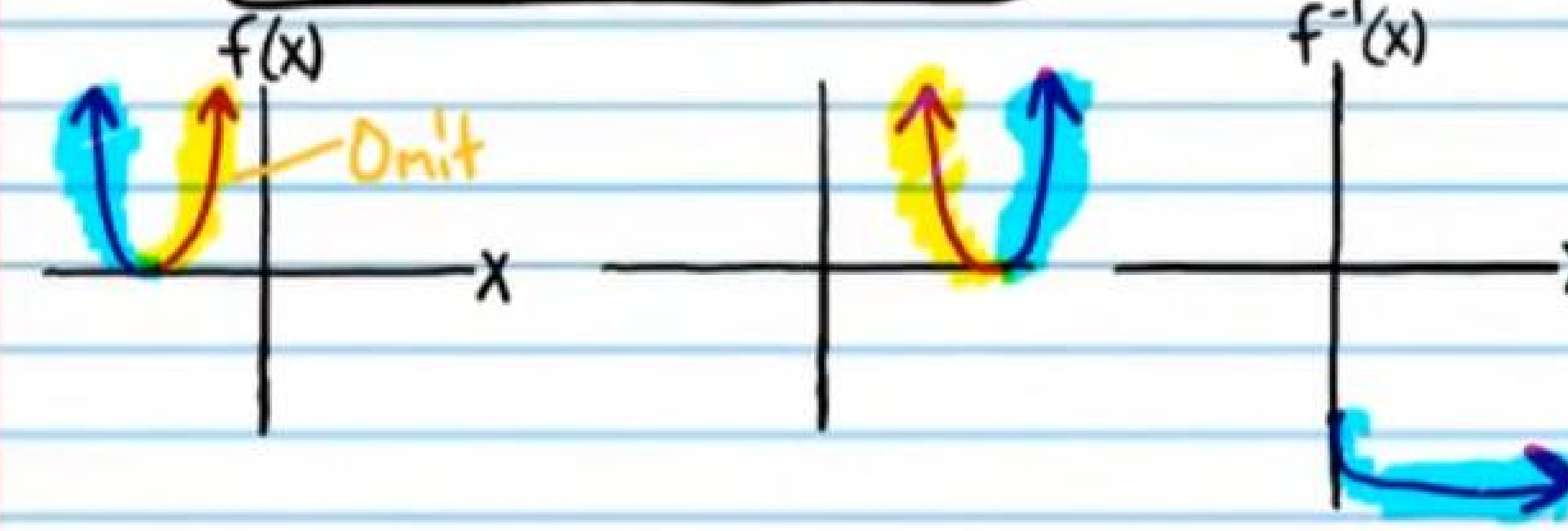
(23) $f(x) = (x+4)^2, \boxed{x \geq -4}$

$\boxed{f^{-1}(x) = \sqrt{x} - 4}$



or $f(x) = (x+4)^2, \boxed{x \leq -4}$

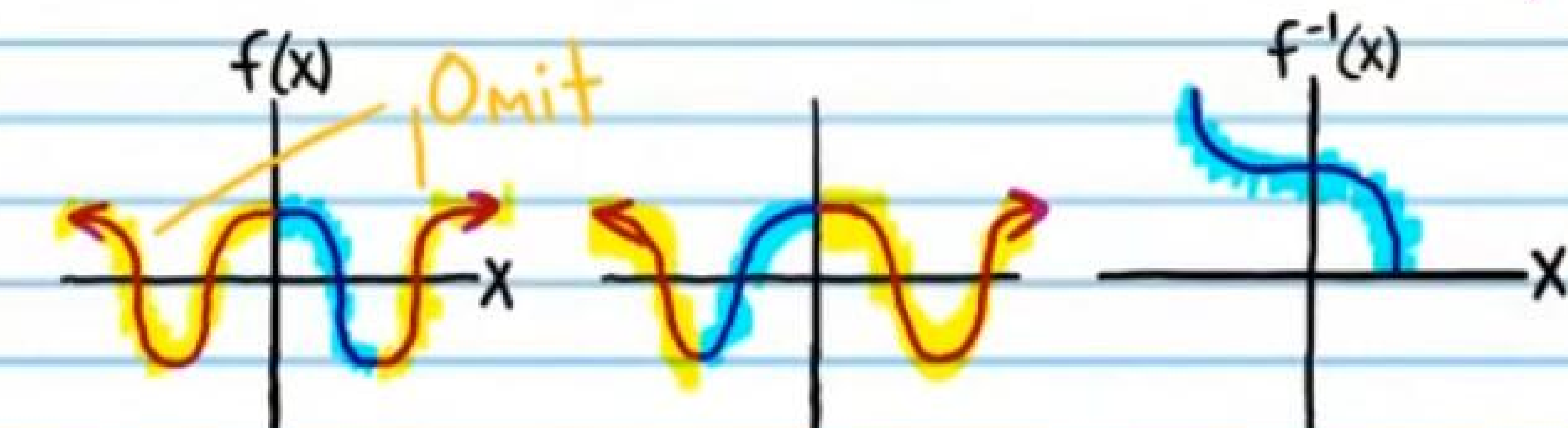
$\boxed{f^{-1}(x) = -\sqrt{x} - 4}$



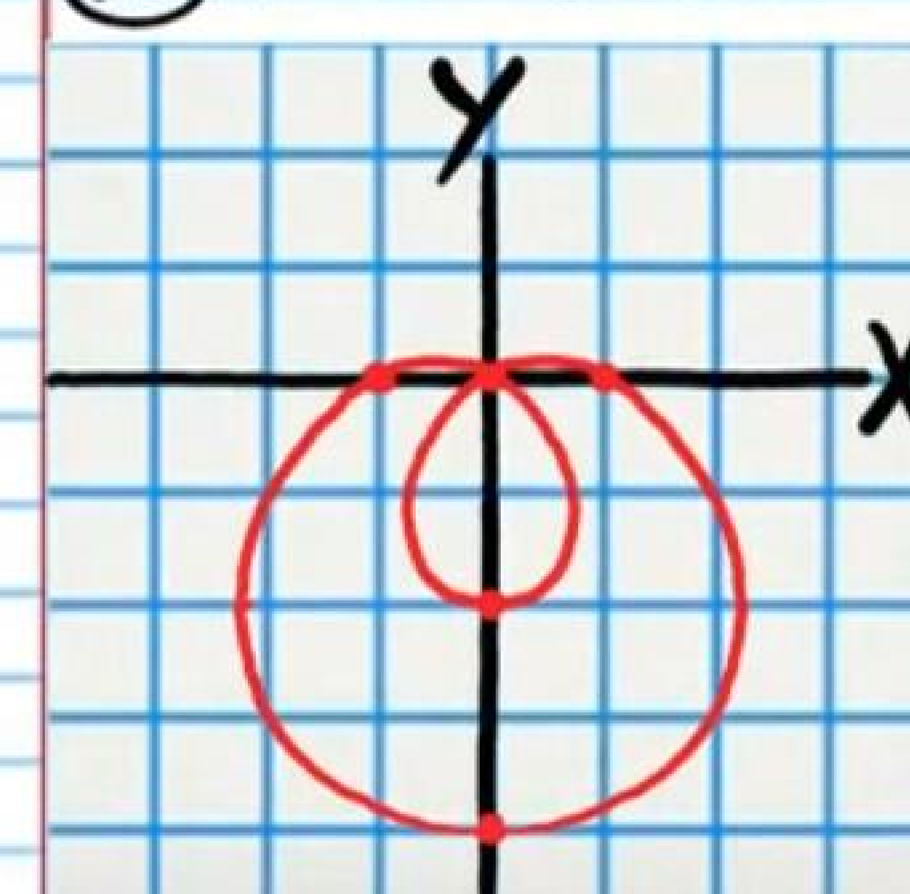
(24) $f(x) = \cos x, \boxed{0 \leq x \leq 180°}$

$\boxed{f^{-1}(x) = \cos^{-1} x,}$ $0 \leq y \leq 180°$

Range restriction is implied and not necessary to write.



(25) $r = 1 - 3\sin\theta$



Other points are possible. Only 4 are necessary.

$(1, 0)$ $(-2, \frac{\pi}{2})$ $(4, -\frac{\pi}{2})$ $(1, \pi)$

$(-\frac{1}{2}, \frac{\pi}{6})$ $(2\frac{1}{2}, -\frac{\pi}{6})$ $(2\frac{1}{2}, -\frac{5\pi}{6})$